Abdullah Al Zahid

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SUMMARY

Transportation engineer with a master's degree whose work has centred on capacity, modelling and data: OpenTrack rail simulation at Metrolinx, agent-based (MATSim) and macroscopic network models in graduate research, and travel and network data analysis at the City of Toronto. Builds the Python tooling and datasets behind the analysis and presents results to the people who act on them.

EXPERIENCE

Assistant Planner, Transportation Planning

City of Toronto | January 2024 – present

- Analyze travel and network data, including several years of the Transportation Tomorrow Survey, to establish where capacity falls short and what that means for future service and investment.
- Built the Transportation Data Dashboard, replacing a report rebuilt by hand each quarter with one validated source the division reports from, and saw the change through to adoption.
- Build and maintain the road and right-of-way datasets the division depends on, using GIS and Python, including a ROW-width dataset covering all city roads, built with QGIS, MapBox and JavaScript.
- Run assigned studies from scoping through to recommendations, writing the briefing notes that come out of them and presenting to planners, engineers and managers in other divisions.

Transportation Planner, Simulation and Modelling

Metrolinx, Service Design | February 2023 – January 2024

- Ran OpenTrack simulations of GO rail services to establish what proposed operating and infrastructure changes would do to run times, reliability and recovery before they were taken forward.
- Built the Crew Variance Tool, a Python application that read large JSON operations files and produced summaries of speed profiles and run-time variance across subdivisions.
- Set up a command line workflow so temporary and permanent slow orders could be pre-simulated in batches, cutting turnaround on scenario requests from days to hours.
- Worked with schedulers, operations staff and engineers on service design decisions and presented the evidence behind them.

Graduate Transportation Planner

Mott MacDonald | January – February 2023

- Short consulting placement on Ontario Line work. Ran traffic simulations to estimate how construction staging would affect surrounding streets, and prepared the supporting material for the project team.

Graduate Research Assistant

University of Alberta | January 2020 – August 2022

- Master's research on how road networks perform during emergency evacuations, using macroscopic (LWR) and agent-based (MATSim) models.
- Worked with large travel datasets and wrote up results for readers without a modelling background.

SKILLS

Simulation: OpenTrack, VISSIM, Synchro and MATSim. Python (Pandas, NumPy) and SQL for analysis and automation. Advanced Excel, including large multi-source workbooks. ArcGIS Pro and QGIS. Dashboards, reports and presentations for technical and non-technical audiences.

EDUCATION

MSc, Transportation Engineering — University of Alberta, 2022

BSc, Civil Engineering — Bangladesh University of Engineering and Technology, 2018

BSc, Computer Science — University of British Columbia, in progress (2025–2027)